

2025-2026 – Econ 0039 – Advanced Macro

Lecture 5: Planned Economies versus Market Economies

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0. Introduction

What is this lecture about?

- ▶ “Market economies” or “Capitalist economies” are ubiquitous (with more or less State intervention).
- ▶ Central planned communist economies are now the exception (China, Cuba, Laos, Vietnam, and North Korea)
- ▶ Formerly communist countries:
 - × Formerly part of the Soviet Union: Armenia, Azerbaijan, Belarus, Estonia, Georgia, Kazakhstan, Kyrgyzstan, Latvia, Lithuania, Moldova, Russia, Tajikistan, Turkmenistan, Ukraine, and Uzbekistan.
 - × Other Asian countries: Afghanistan, Cambodia, Mongolia, and Yemen.
 - × Soviet-controlled Eastern bloc countries: Bulgaria, Czech Republic, Germany (East), Hungary, Poland, Romania, Slovakia.
 - × The Balkans: Albania, Bosnia and Herzegovina, Bulgaria, Croatia, Rep. of Macedonia, Montenegro, Serbia, and Slovenia.
 - × Africa: Angola, Benin, Dem Rep. of Congo, Ethiopia, Somalia, Eritrea, and Mozambique.

0. Introduction

What is this lecture about?

- ▶ Beyond politics, what does macroeconomics tell us about the relative merits of the two systems?
- ▶ We will discuss history of thought, study a simple model and then look at one particular historical episode: the USSR.

Plan of the Lecture

1. The Economic Calculation Debate
2. A Model of the Trade-off Between Markets and Planning
3. Central Planning in Action: The USSR
4. An Account of Russian Economic Development from 1885 to 1940

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1. The Economic Calculation Debate

Question

- ▶ Capitalism: economic decision making is done by individuals largely in their roles as consumers, owners of factors of production, and managers of firms; most economic resources are privately owned.
- ▶ Socialism: economic decision making is done by individuals largely in their roles as voters, politicians, and managers of firms; economic resources may be privately or publicly owned, but the control over resource allocation is by government, not by the owners of the resources.
- ▶ How would decisions be made *in practice* in a socialist (centrally planned) economy ? $\rightsquigarrow$ *Economic Calculation debate* (a.k.a. *Socialist Calculation debate*)
- ▶ MARX is silent about this question.
- ▶ A theoretical question that became a practical one after the Russian revolution of 1917.

1. The Economic Calculation Debate

The two fundamental theorems of Welfare Economics

- ▶ First theorem: Competitive equilibrium (with complete markets and perfect information) is Pareto optimal.
- ▶ Second theorem: Any Pareto optimum can be supported as a competitive equilibrium for some initial set of endowments.
- ▶ Rigorously formulated by PAUL A SAMUELSON (1947)

1. The Economic Calculation Debate

Markets

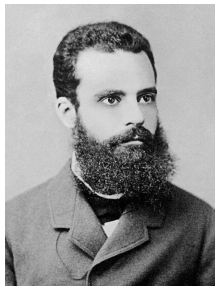
- ▶ Since ADAM SMITH, we understand how, in principle, competitive markets can efficiently organise production and trade (invisible hand $\approx$ first theorem of welfare).
- ▶ Prices are important as indicating relative scarcity of goods and factors of production.

1. The Economic Calculation Debate

Planning

- ▶ VILFREDO PARETO, "*Les systèmes socialistes*", (1902-03)
- ▶ Central planning = solving for a Planner problem $\rightsquigarrow$ no reason why maximum welfare could not be achieved under socialism.

Figure 1: PARETO



1. The Economic Calculation Debate

Planning

- ▶ ENRICO BARONE, "*The Ministry of Production in a Collectivist State*", 1908
 - × First shows the conditions that lead to maximum welfare under capitalism with perfectly competitive markets
 - × Then builds a model of socialism:
 - ▶ all resources other than labor are collectively owned
 - ▶ a ministry of production controls the economy
 - × If the ministry set prices so that they are equal to costs of production and if costs of production are at a minimum, an optimum allocation of resources exists and maximum welfare is achieved.
- ▶ The problem seems to be theoretically solved: socialism can do as well as capitalism.

Figure 2: BARONE



1. The Economic Calculation Debate

LUDWIG VON MISES

- ▶ LUDWIG VON MISES, *"Economic Calculation in a Socialist Commonwealth"*, (1920)

"Since recent events helped socialist parties to obtain power in Russia, Hungary, Germany and Austria, and have thus made the execution of a socialist nationalization program a topical issue, Marxist writers have themselves begun to deal more closely with the problems of the regulation of the socialist commonwealth. But even now they still cautiously avoid the crucial question, leaving it to be tackled by the despised "Utopians". They themselves prefer to confine their attention to what is to be done in the immediate future; they are forever drawing up programs of the path to Socialism and not of Socialism itself. The only possible conclusion from all these writings is that they are not even conscious of the larger problem of economic calculation in a socialist society."

Figure 3: MISES



1. The Economic Calculation Debate

LUDWIG VON MISES

- ▶ *“Every step that takes us away from private ownership of the means of production and from the use of money also takes us away from rational economics.”* MISES (1920)
- ▶ For MISES, rational allocation of resources is impossible under socialism.
- ▶ Key role played by factor markets: In these markets owners of land, labor, and capital supply the factors of production to firms demanding them. Prices emerge, and on the basis of these prices and the available technology, firms make decisions on the most economical way to combine factors to produce final products.
- ▶ Under socialism the factors of production are, for the most part, not individually owned; they are owned by the community.
- ▶ Because there are no independent owners of the factors of production, there would be no factor markets and no set of prices emerging from these markets. Rational decision making for resource allocation is not possible without factor prices
- ▶ Not true in theory, as already shown by PARETO and BARONE

1. The Economic Calculation Debate

FRIEDRICH AUGUST VON HAYEK and LIONEL ROBBINS

- ▶ HAYEK and ROBBINS agree that the solution to the allocation problem under socialism is theoretically possible.
- ▶ But they claim it is *practically* impossible:
 - × The Socialist Planner does not have enough information,
 - × Even if they had, it would be impossible to solve for a gigantic system of simultaneous equations.

Figure 4: HAYEK & ROBBINS



1. The Economic Calculation Debate

OSKAR LANGE

- ▶ OSKAR LANGE, "*On the Economic Theory of Socialism*", 1936-37
- ▶ Under competitive capitalism, neoclassical theory has found that three conditions obtain in equilibrium.
 - × (1) Both consumers and producers are in maximizing positions;
 - ▶ (a) consumers are spending their limited income so as to maximize satisfaction,
 - ▶ (b) producers are maximizing profits.
 - × (2) Every price is such that quantity supplied equals quantity demanded, so that all markets are cleared.
 - × (3) Incomes from consumers will be equal to their receipts from the factors sold plus profits.

Figure 5: LANGE



1. The Economic Calculation Debate

OSKAR LANGE

- ▶ Under planned socialism equilibrium
 - × Condition (1a) is unchanged
 - × Condition (1b) no longer holds under socialism $\rightsquigarrow$ replace by requiring producers to
 - ▶ produce every output at the lowest possible cost;
 - ▶ choose the scale of output so that price equals marginal cost.
 - × Condition (2) is not achieved by free-market forces, but by State Planners adjusting prices on a *trial-and-error* basis.
 - × Condition (3) would hold under socialism, except that there would be no profits.
- ▶ The *trial-and-error* mechanism implies that there is no more need for a huge computer solving supply-and-demand equations under socialism than under capitalism.

1. The Economic Calculation Debate

A quantitative question

- ▶ In theory, a Planner can do as well as the market.
- ▶ In practice, information is key, and the Planner may not have all the information, so that market is preferred.
- ▶ But lack of information is also pervasive in market economies:
 - × Externalities
 - × Managers & politicians incentives.
 - × Goods quality known to the seller, not the buyer.
- ▶ In the end it is a quantitative question that can be answered by modelling and looking at historical episodes.
- ▶ Very likely that the best is some intermediate solution between markets and planning.
- ▶ Remark: all this disregards the political dimension.
- ▶ There are example of democratic planning (*Dirigism* in the UK and in France after WWII) (de Gaulle “*The plan is a burning obligation*”)

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2. A Model of the Trade-off Between Markets and Planning

Pros and cons

Market

► Pros

- × Choices are made by firms/hh who know their technology/preferences

► Cons

- × Perfect competition is needed for optimality of equilibrium
- × (also complete markets)
- × Externalities
- × Asymmetries of information $\rightsquigarrow$ missing markets (unemployment insurance)

Planning

► Pros

- × Externalities can be internalized by the Planner

► Cons

- × Planner can be non benevolent
- × Information might be incomplete
- × Asymmetries of information $\rightsquigarrow$ moral hazard

2. A Model of the Trade-off Between Markets and Planning

- ▶ Let's write a model in which one can generate one of the trade-offs.
- ▶ Key trade-off in the model
 - × *Information is complete* in the market solution, but *imperfect competition*
 - × *Information is incomplete* in the planning solution, but *no monopoly distortions*

2. A Model of the Trade-off Between Markets and Planning

Preference and Technology

- ▶ Static economy, one good
- ▶ One representative household

$$U(C, L) = C^\beta - BL,$$

C is consumption, L is labour, $B > 0$, $\beta \in]0, 1[$

- ▶ Technology

$$C = AL^\alpha,$$

$$A > 0, \alpha \in]0, 1[$$

2. A Model of the Trade-off Between Markets and Planning

Frictionless Planner problem

$$\begin{array}{ll}\max_{C,L} & C^\beta - BL \\ \text{s.t.} & C = AL^\alpha\end{array}$$

- Rewrite as

$$\max_L (AL^\alpha)^\beta - BL$$

- FOC:

$$\alpha\beta A^\beta L^{\alpha\beta-1} = B$$

- Solution (FB stands for “First Best”)

$$C^{FB} = \left(\frac{\alpha\beta}{B}\right)^{\frac{\alpha}{1-\alpha\beta}} A^{\frac{1}{1-\alpha\beta}}$$

$$L^{FB} = \left(\frac{\alpha\beta}{B}\right)^{\frac{1}{1-\alpha\beta}} A^{\frac{\beta}{1-\alpha\beta}}$$

2. A Model of the Trade-off Between Markets and Planning

Frictionless Planner problem

- We therefore have

$$U^{FB} = (1 - \alpha\beta)(\alpha\beta)^{\frac{\alpha\beta}{1-\alpha\beta}} A^{\frac{\beta}{1-\alpha\beta}} B^{\frac{-\alpha\beta}{1-\alpha\beta}}$$

- Check that $\frac{\partial U^{FB}}{\partial A} > 0$ and $\frac{\partial U^{FB}}{\partial B} < 0$.

2. A Model of the Trade-off Between Markets and Planning

Competitive equilibrium

- ▶ Competitive equilibrium is (C, L, Π) and (p, w) such that
 - × given Π , p and w , household chooses optimally C and L ,
 - × given p and w , Π is maximised by the representative firm,
 - × p, w are such that good and labour markets clear.

2. A Model of the Trade-off Between Markets and Planning

Competitive equilibrium

- ▶ Household consumes, receives labour income and representative firm profit

$$\begin{aligned} \max_{C,L} \quad & C^\beta - BL \\ \text{s.t.} \quad & pC = wL + \Pi \end{aligned}$$

- ▶ Replace L using budget constraint and obtain FOC

$$\beta C^{\beta-1} - \frac{Bp}{w} = 0$$

which gives good demand

$$C = \left(\frac{Bp}{\beta w} \right)^{-\frac{1}{1-\beta}}$$

and labour supply is obtained using the BC.

- ▶ Note that good demand rewrite

$$\frac{w}{p} = \frac{B}{\beta} C^{1-\beta} \quad (\star)$$

2. A Model of the Trade-off Between Markets and Planning

Competitive equilibrium

- Representative firm

$$\begin{array}{ll} \max_{C,L} & \Pi = pC - wL \\ \text{s.t.} & C = AL^\beta \end{array}$$

- Substitute L in Π and take FOC wrt C to obtain

$$C = A^{\frac{1}{1-\alpha}} \alpha^{\frac{\alpha}{1-\alpha}} \left(\frac{w}{p} \right)^{-\frac{\alpha}{1-\alpha}} \quad (**)$$

2. A Model of the Trade-off Between Markets and Planning

Competitive equilibrium

- Equilibrium: Good market equilibrium is obtained by taking w/p in $(\star)$ and substitute in $(\star\star)$ to obtain:

$$C^{CE} = \left(\frac{\alpha\beta}{B} \right)^{\frac{\alpha}{1-\alpha\beta}} A^{\frac{1}{1-\alpha\beta}}$$

where CE stands for “Competitive Equilibrium”

- Recall that

$$C^{FB} = \left(\frac{\alpha\beta}{B} \right)^{\frac{\alpha}{1-\alpha\beta}} A^{\frac{1}{1-\alpha\beta}}$$

- Note that $C^{CE} = C^{FB}$, $L^{CE} = L^{FB}$ and $U^{CE} = U^{FB}$
- This is the first Welfare theorem.

2. A Model of the Trade-off Between Markets and Planning

Introducing Frictions

- ▶ Absent of frictions, a (socialist) Planner does as well as the (capitalist) competitive equilibrium.
- ▶ Now introduce frictions:
 - × The market economy is not perfectly competitive
 - × The Planner does not have perfect information on the economy
- ▶ Let's compute utility in these two situations and compare.

2. A Model of the Trade-off Between Markets and Planning

Market economy with monopoly power

- ▶ Assume now that there is one monopoly in the economy, maximising profit knowing good demand function.
- ▶ Household problem is as before.

$$\begin{array}{ll}\max_{C,L} & C^\beta - BL \\ \text{s.t.} & pC = wL + \Pi\end{array}$$

which gives good demand

$$C = \left(\frac{Bp}{\beta w} \right)^{-\frac{1}{1-\beta}}$$

- ▶ Price elasticity of demand is $-\frac{1}{1-\beta}$
- ▶ When $\beta \rightarrow 1$, demand elasticity $\rightarrow -\infty \rightsquigarrow$ perfect competition.

2. A Model of the Trade-off Between Markets and Planning

Market economy with monopoly power

- The monopoly takes the demand curve into consideration when maximising profit

$$\begin{aligned} \max_{C,L,p} \quad & \Pi = pC - wL \\ \text{s.t.} \quad & C = AL^\beta \\ & C = \left(\frac{Bp}{\beta w} \right)^{-\frac{1}{1-\beta}} \end{aligned}$$

- Profit then rewrite

$$\Pi = \frac{\beta w A^\beta}{B} L^{\alpha\beta} - wL$$

and FOC implies in equilibrium (IC stands for “Imperfect Competition”)

$$L^{IC} = \left(\beta(\alpha\beta) \frac{A^\beta}{B} \right)^{\frac{1}{1-\alpha\beta}} = \beta^{\frac{1}{1-\alpha\beta}} L^{FB} < L^{FB}$$

so that

$$C^{IC} = \beta^{\frac{\alpha}{1-\alpha\beta}} C^{FB} < C^{FB}$$

2. A Model of the Trade-off Between Markets and Planning

Market economy with monopoly power

- We can also compute welfare

$$U^{IC} = (1 - \alpha\beta^2) (\alpha\beta^2)^{\frac{\alpha\beta}{1-\alpha\beta}} A^{\frac{\beta}{1-\alpha\beta}} B^{-\frac{\alpha\beta}{1-\alpha\beta}}$$

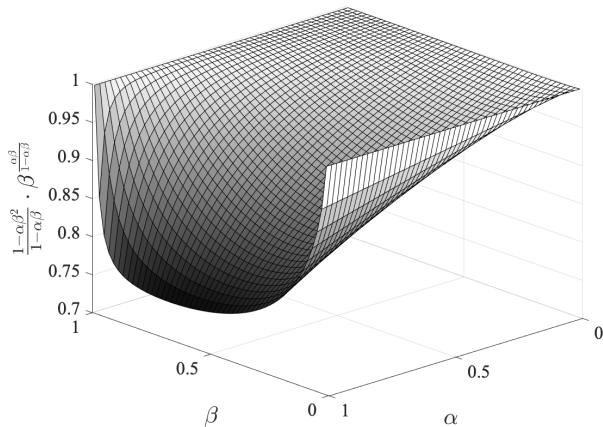
and

$$\frac{U^{IC}}{U^{FB}} = \underbrace{\frac{(1 - \alpha\beta^2)}{(1 - \alpha\beta)}}_{>1} \times \underbrace{\beta^{\frac{\alpha\beta}{1-\alpha\beta}}}_{<1} < 1$$

- Monopoly power creates distortion (price above marginal cost and therefore under-production) and leads to inefficient outcomes.
- Note that when $\beta \rightarrow 1$, price elasticity of demand $\rightarrow \infty$, so that monopoly distortion vanishes and $\frac{U^{IC}}{U^{FB}} \rightarrow 1$.

2. A Model of the Trade-off Between Markets and Planning

Market economy with monopoly power: plotting $\frac{U^{IC}}{U^{FB}}$



2. A Model of the Trade-off Between Markets and Planning

Uncertainty

- Assume that B is random, and follows a log-normal distribution:

$$\log B \rightsquigarrow \mathcal{N}(\log \bar{B}, \sigma^2)$$

- Note that

$$E[B^n] = \bar{B}^n e^{\frac{n^2}{2}\sigma^2}$$

2. A Model of the Trade-off Between Markets and Planning

Planning problem with imperfect information

- ▶ Assume that the Planner does not observe B , and has to make production and consumption plans.
- ▶ The Planner will maximise expected utility:

$$\begin{aligned} \max_{C,L} \quad & E [C^\beta - BL] \\ \text{s.t.} \quad & C = AL^\alpha \end{aligned}$$

× Rewrite as

$$\max_L (AL^\alpha)^\beta - E[B]L$$

× Solution (II stands for “Imperfect Information”)

$$C'' = \left(\frac{\alpha\beta}{E[B]} \right)^{\frac{\alpha}{1-\alpha\beta}} A^{\frac{1}{1-\alpha\beta}}$$

$$L'' = \left(\frac{\alpha\beta}{E[B]} \right)^{\frac{1}{1-\alpha\beta}} A^{\frac{\beta}{1-\alpha\beta}}$$

and $E[B] = \bar{B}$

2. A Model of the Trade-off Between Markets and Planning

Planning problem with imperfect information

► Recall

$$U^{FB} = (1 - \alpha\beta)(\alpha\beta)^{\frac{\alpha\beta}{1-\alpha\beta}} A^{\frac{\beta}{1-\alpha\beta}} B^{\frac{-\alpha\beta}{1-\alpha\beta}}$$

► We have therefore welfare under planning with imperfect information :

$$\begin{aligned} E[U^H] &= (1 - \alpha\beta)(\alpha\beta)^{\frac{\alpha\beta}{1-\alpha\beta}} A^{\frac{\beta}{1-\alpha\beta}} (E[B])^{-\frac{\alpha\beta}{1-\alpha\beta}} \\ &= (1 - \alpha\beta)(\alpha\beta)^{\frac{\alpha\beta}{1-\alpha\beta}} A^{\frac{\beta}{1-\alpha\beta}} \bar{B}^{-\frac{\alpha\beta}{1-\alpha\beta}} \end{aligned}$$

2. A Model of the Trade-off Between Markets and Planning

Market economy with monopoly power

- ▶ Assume the monopoly observes B *before* setting its price (perfect information).
- ▶ The monopoly behaviour is as before.
- ▶ Recall

$$U^{IC} = (1 - \alpha\beta^2) (\alpha\beta^2)^{\frac{\alpha\beta}{1-\alpha\beta}} A^{\frac{\beta}{1-\alpha\beta}} B^{-\frac{\alpha\beta}{1-\alpha\beta}}$$

- ▶ Expected welfare in the monopoly economy will be

$$\begin{aligned} E[U^{IC}] &= (1 - \alpha\beta^2) (\alpha\beta^2)^{\frac{\alpha\beta}{1-\alpha\beta}} A^{\frac{\beta}{1-\alpha\beta}} E\left[B^{-\frac{\alpha\beta}{1-\alpha\beta}}\right] \\ &= (1 - \alpha\beta^2) (\alpha\beta^2)^{\frac{\alpha\beta}{1-\alpha\beta}} A^{\frac{\beta}{1-\alpha\beta}} \overline{B}^{-\frac{\alpha\beta}{1-\alpha\beta}} e^{\frac{1}{2} \frac{(\alpha\beta)^2}{(1-\alpha\beta)^2} \sigma^2} \end{aligned}$$

2. A Model of the Trade-off Between Markets and Planning

Comparing market and planning allocations with imperfect competition/information

- We have therefore welfare under planning with imperfect information or under Imperfect Competition:

$$\begin{aligned} E[U^I] &= (1 - \alpha\beta)(\alpha\beta)^{\frac{\alpha\beta}{1-\alpha\beta}} A^{\frac{\beta}{1-\alpha\beta}} \bar{B}^{-\frac{\alpha\beta}{1-\alpha\beta}} \\ E[U^{IC}] &= (1 - \alpha\beta^2)(\alpha\beta^2)^{\frac{\alpha\beta}{1-\alpha\beta}} A^{\frac{\beta}{1-\alpha\beta}} \bar{B}^{-\frac{\alpha\beta}{1-\alpha\beta}} e^{\frac{1}{2} \frac{(\alpha\beta)^2}{(1-\alpha\beta)^2} \sigma^2} \end{aligned}$$

2. A Model of the Trade-off Between Markets and Planning

Comparing market and planning allocations with imperfect competition/information

$$\frac{E[U^{IC}]}{E[U^I]} = \underbrace{\left[\frac{1 - \alpha\beta^2}{1 - \alpha\beta} \beta^{\frac{\alpha\beta}{1-\alpha\beta}} \right]}_{\mathcal{A}} \times \underbrace{\frac{E \left[B^{-\frac{\alpha\beta}{1-\alpha\beta}} \right]}{E[B]^{-\frac{\alpha\beta}{1-\alpha\beta}}}}_{\mathcal{B}}$$

- ▶ $\mathcal{A}$ is the distortion related to imperfect competition: $\mathcal{A} < 1$ and $\mathcal{A} \rightarrow 1$ when $\beta \rightarrow 1$ (perfect competition)
- ▶ $\mathcal{B}$ is the distortion related to imperfect information of the Planner.
 - × $\mathcal{B} = e^{\frac{1}{2} \frac{\alpha\beta}{(1-\alpha\beta)^2} \sigma^2}$
 - × $\mathcal{B} > 1$ and $\mathcal{B} \rightarrow 1$ when $\sigma^2 \rightarrow 0$ (perfect information)
- ▶ $\mathcal{A} < 1$ and $\mathcal{B} > 1$: which social organisation is preferred depends on the relative size of the two distortions.

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3. Central Planning in Action: The USSR

Geography



3. Central Planning in Action: The USSR

Short history

- ▶ Before 1917: Russian Empire (Emperor Nicholas II, first cousin of UK King George V and of Wilhelm II of Germany)
- ▶ 1917: Bolshevik revolution (Vladimir Ilyich Ulyanov a.k.a. Lenin)
- ▶ “Union of Soviet Socialist Republics”. (CCCP in Russian):
 - × One-party State (Communist Party of the Soviet Union)
 - × “Democratic centralism” (“*freedom of discussion, unity of action*”)
 - × Almost all sectors of the economy are State-owned.
 - × Central planning (5-year plans) organises the economy after 1928
- ▶ 1917-1923: Civil war between Reds and Whites

Nicholas II



Lenin



3. Central Planning in Action: The USSR

Short history

- ▶ 1928-1953: Stalin is General Secretary of the Communist Party of the Soviet Union
- ▶ Political terror under Stalin
 - × Soviet famine of 1932-33 caused 7.9 millions deaths (including *Holodomor* in Ukraine)
 - × Gulag (forced labour camps) 19 millions from 1922 to 1953
 - × 1 million political executions
- ▶ After WWII: Cold war
- ▶ 1953-1964: De-Stalinization (Khrushchev)
- ▶ 1964-1985: Sclerosis with Brezhnev, Andropov and Chernenko
- ▶ 1985-1991: Gorbachev: Perestroika (reconstruction) and Glasnost (transparency) reforms.
- ▶ 1989: fall of the Berlin wall $\rightsquigarrow$ 1990-1991: dissolution and collapse of the USSR.

Stalin



Khrushchev



Brezhnev

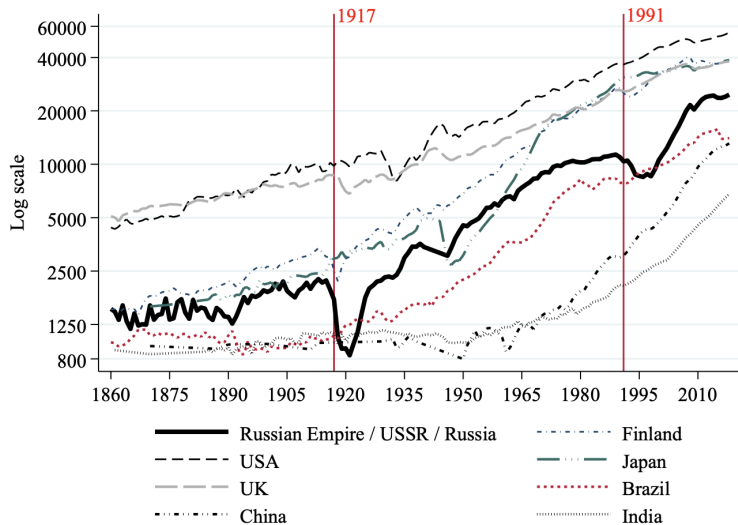


Gorbachev



3. Central Planning in Action: The USSR

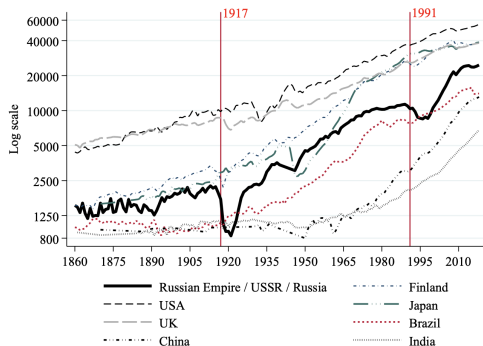
GDP Per Capita in Russia and Selected Countries, 2011 international dollars



3. Central Planning in Action: The USSR

Short economic history

- ▶ 1860: Russian income per capita is
 - × 1/3 of UK or USA
 - × 2 × India, China
 - × $\approx$ Japan, Finland
- ▶ Important reform in 1861: Emancipation of serfs
- ▶ From 1890's to WWI: structural transformation from agriculture to manufacturing.
- ▶ But in 1913, still +50% of GDP in agric.
- ▶ In 1913, Russian income p.c. is 22% of USA, 30% of UK.

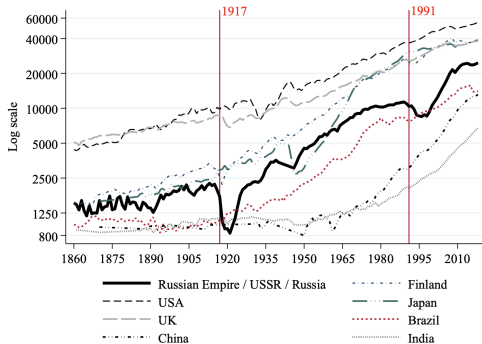


Source: Zhuravskaya et al. (2022))

3. Central Planning in Action: The USSR

Short economic history

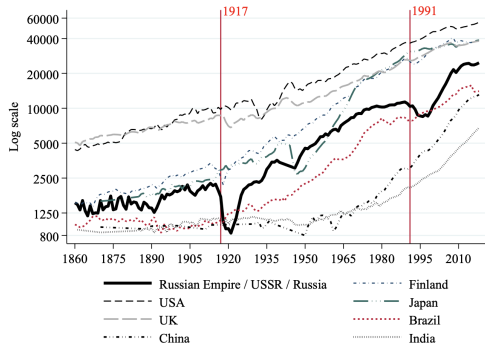
- ▶ 1914: WWI + 1917: Bolshevik revolution (Lenin) + Civil war
- ▶ ~→ Economic disaster (famine of 1921 killed 5 millions)
- ▶ New Economic Policy (NEP) from 1921 to 1928: reintroduction of a limited market economy. (“free market and capitalism, both subject to state control” (LENIN [1922]))



3. Central Planning in Action: The USSR

Short economic history

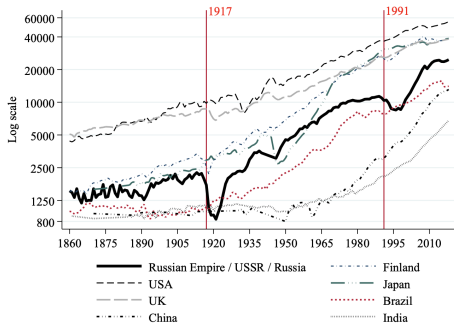
- ▶ 1928-1953: Stalin
- ▶ From 1928, massive reallocation of labour from agriculture to manufacturing + collectivisation of agriculture.
- ▶ $\rightsquigarrow$ Sustained growth until WWII (“the Great Patriotic War”)



3. Central Planning in Action: The USSR

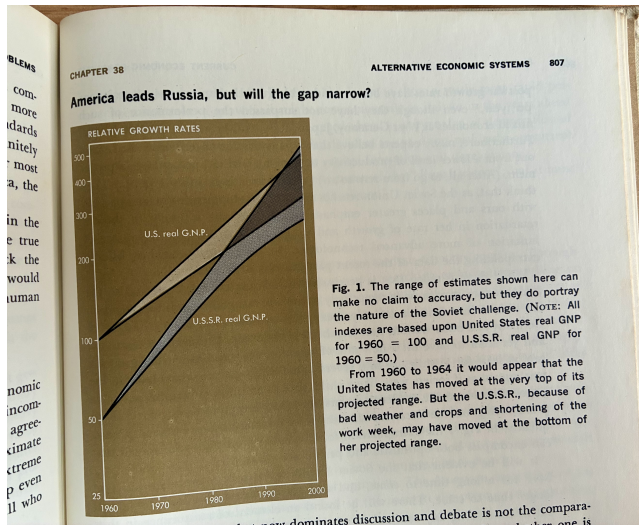
Short economic history

- ▶ From late 1940's to 1991: USSR is an *upper middle income country* (World Bank definition)
- ▶ Late 1940's to early 70's: sustained growth, parallel to the US (but no catchup as Finland or Japan)
 - × growth above 3% per year
 - × challenging the USA : first satellite - Sputnik (1957), first man in space - Yuri Gagarin (1961)
- ▶ 1970's and 1980's: stagnation (0.7% of gdp p.c. in the 1980's)
- ▶ Central planing efficient to allocate resources towards investment in the 50-60's, not for the development of a mass consumption economy.



3. Central Planning in Action: The USSR

"Economics" Textbook, SAMUELSON, 1964 edition



3. Central Planning in Action: The USSR

Organization of planning

- ▶ For every good, the plan aims at estimating a *material balance*:

$$\begin{aligned} \text{production} + \text{imports} &= \text{intermediate consumption} + \text{final cons.} + \text{exports} \\ &+ \text{change in inventories} \end{aligned}$$

- ▶ The GOSPLAN (State Planning Committee) designs material balances for 2000 aggregate product groups
- ▶ The GOSNAB does the allocation of producer goods (disaggregated into 15,000 categories) to enterprises.
- ▶ Industrial branch ministries disaggregate into 50,000 categories of 10 to 15 goods
~> a total of 500,000 to 750,000 goods (out of 25 millions of goods)
- ▶ When material balances does not add up: bargaining between the enterprises and the various level of planning authorities

3. Central Planning in Action: The USSR

GOSPLAN

- ▶ GOSPLAN's must implement politically determined objectives by designing a consistent set of plan targets.
- ▶ Need for coherence between production and supply at all times, but also between investment plans and the current production of capital goods, and also take account of foreign trade.
- ▶ Successive versions of the plan are revised until a general balance was attained.
- ▶ No decentralisation in theory: the coordination of all these decisions has to be consciously achieved by the Planners.

3. Central Planning in Action: The USSR

Inefficiencies

- ▶ In practice, the actual plans are aggregated (e.g., tons of metal, millions of square metres of cloth, millions of rubles' worth of construction or of furniture), so that decisions on the product mix are necessarily decentralized.
- ▶ A lot of inefficient decisions:
 - × Quality is sacrificed in order to fulfil the plan in quantitative terms.
 - × "Plan-fulfilment" is the only objective $\rightsquigarrow$ managers hide their true production potential.
 - × Planned targets expressed in tons encourages excessive weight in the product concerned,
 - × Targets expressed in rubles discourages economy and rewards the use of expensive materials.
 - × Prices do not reflect supply and demand.
- ▶ Shortages and queues

3. Central Planning in Action: The USSR

In the *Pravda*



In a letter to the editors of Pravda, the deputy director of the Riazan heavy machine-building factory complains of the lack of coordination between output plans and planned supplies—of raw materials, semifabricates, and parts. He cites an example from his own factory.

This year, the regional sales-supply organ for electrical manufactures allotted to the Riazan factory only half the number of a particular electric coupling that it needed. The factory, in difficulty and anxious, turned first to its regional Economic Council, which explained, “calmly,” that it and the sales-supply organ were in disagreement on the allocation. This was in mid January, when the annual plan had already been approved in detail.

3. Central Planning in Action: The USSR

In the *Pravda*

Finally the officials of the Economic Council got it through their heads that this was a serious business. "Phone calls and telegrams went out, as is usual in such cases." And then, on the 2nd of February, the director of the sales-supply organ received a group of workers from the factory. He told them, however, that nothing could be done: he lacked the resources. Nor would he agree to deliver their 3rd and 4th quarter allotments in the first two quarters. In the end, he "leveled at" them "such astronomic figures" as almost to convince them their plight was normal.

Subsequently, they approached the Economic Council of the Russian Republic, the central sales-supply organ for electrical manufactures, attached to Gosplan USSR, and many other institutions. No one helped.

"When, finally, it became clear that no one was going to help us, we sent our representatives to Kharkov, then our workmen, and, finally, two trucks, which, spelling one another, made several dozen trips over the Riazan-Kharkov road, transporting couplings in small batches. Of course, each coupling cost us 'a large kopek.' But we had no other way out. All the same, the factory fulfilled its plan for the first quarter. But what a strain it cost us!"

Pravda, May 4, 1961 (Powell (1977))

3. Central Planning in Action: The USSR

Shortages (classic Soviet joke)

- *“I want to sign up for the waiting list for a car. How long is it?”*
- *“Exactly ten years from today.”*
- *“Morning or afternoon?”*
- *“Why, what difference does it make?”*

3. Central Planning in Action: The USSR

Shortages (classic Soviet joke)

- *“I want to sign up for the waiting list for a car. How long is it?”*
- *“Exactly ten years from today.”*
- *“Morning or afternoon?”*
- *“Why, what difference does it make?”*
- *“Well, the plumber is due in the morning.”*

Plan of the Lecture

1. The Economic Calculation Debate
2. A Model of the Trade-off Between Markets and Planning
3. Central Planning in Action: The USSR
4. An Account of Russian Economic Development from 1885 to 1940

4. An Account of Russian Economic Development from 1885 to 1940

The question

- ▶ Based on “*The Industrialization and Economic Development of Russia through the Lens of a Neoclassical Growth Model*”, Cheremukin et al. (2016)
- ▶ Take a Neoclassical two-sector model (agriculture and manufacturing), generally used to study structural transformation.
- ▶ Actual data can be seen as generated by competitive equilibrium (or frictionless Planner), distorted by some *wedges*.
- ▶ These wedges tell us where the inefficiencies were.
- ▶ Two sub-periods are studied
 - × The Tsarist Russia industrialisation (1885-1913)
 - × The pre-war Stalin industrialisation (1928-1940)

4. An Account of Russian Economic Development from 1885 to 1940

The Tsarist Russia industrialisation (1885-1913)

- ▶ Abolition of serfdom in 1861: Russian peasants received freedom and land rights in exchange for redemption payments.
- ▶ The land was given to communal property of villages rather than transferred to private property of individual households $\rightsquigarrow$ limits to labour mobility.
- ▶ Russian industry is small with significant barriers to entry and widespread monopolies.
- ▶ Russian tsars traditionally distrust capitalist institutions seeing them as a threat to their absolute power.
- ▶ Russian companies were also protected from competition by trade policy imposing high tariffs on non-agricultural imports.

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The pre-war Stalin industrialisation (1928-1940)

- ▶ Stalin followed a policy of expropriation of agricultural surplus in order to finance industrialization.
- ▶ Villages received quotas for grain procurement at below-market prices, with a higher burden falling disproportionately on more prosperous peasants, the *kulaks*.
- ▶ By the early 1930s, the Soviet government had attempted to socialize all agricultural livestock and ban private agricultural markets.
- ▶ Peasants were forced to join newly formed collectives and responded with widespread slaughtering of livestock.
- ▶ Industrialisation is driven by 5-year plans.
- ▶ Main goal of the economic strategy is to overtake the advanced capitalist economies in industrial output per head as quickly as possible.
- ▶ The Soviet government nationalised trade, eliminated the remaining private industry, introduced price controls, and rationed consumer goods.

4. An Account of Russian Economic Development from 1885 to 1940

Model

- Preferences:

$$\sum_{t=0}^{\infty} \beta^t \frac{U(c_t^A, c_t^M)^{1-\rho} - 1}{1-\rho}$$

with

$$U(c_t^A, c_t^M) = \left[\eta^{\frac{1}{\sigma}} \left(c_t^A - \gamma^A \right)^{\frac{\sigma-1}{\sigma}} + (1-\eta)^{\frac{1}{\sigma}} \left(c_t^M \right)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$$

- Preferences are non-homothetic because of the $\gamma^A \rightsquigarrow$ agriculture good is an inferior good, whose share in total expenditures decreases with income.

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Model

► Technology

$$Y_t^A = F^A(K_t^A, N_t^A, L_t^A; X_t^A) = X_t^A \left(K_t^A\right)^{\alpha_{K,A}} \left(N_t^A\right)^{\alpha_{N,A}} \left(L_t^A\right)^{1-\alpha_{K,A}-\alpha_{N,A}}$$

$$Y_t^M = F^M(K_t^M, N_t^M, L_t^M; X_t^M) = X_t^M \left(K_t^M\right)^{\alpha_{K,M}} \left(N_t^M\right)^{\alpha_{N,M}} \left(L_t^M\right)^{1-\alpha_{K,M}-\alpha_{N,M}}$$

where X is Total Factor productivity, K capital and N labour.

- L is land, in fixed supply and assumed to be constant (and =1) in each sector.
- Capital accumulates according to

$$K_{t+1} = (1 - \delta)K_t + I_t$$

with $K_t = K_t^A + K_t^M$.

4. An Account of Russian Economic Development from 1885 to 1940

Model

- Resource constraints:

$$C_t^A + E_t^A = Y_t^A$$

$$C_t^M + I_t + E_t^M = Y_t^M$$

where E^A and E^M are net exports.

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Optimal allocations

► Competitive equilibrium or Planner solution will satisfy:

× Optimal allocation of labour

$$1 = \frac{U_{M,t}}{U_{A,t}} \frac{F_{N,t}^M}{F_{N,t}^A}$$

× Optimal allocation of capital

$$1 = \frac{U_{M,t}}{U_{A,t}} \frac{F_{K,t}^M}{F_{K,t}^A}$$

× Optimal allocation of consumption/savings

$$1 = (1 + F_{K,t+1}^M - \delta) \beta \frac{U_{M,t+1}}{U_{M,t}}$$

Note: F_X^S is the derivative of the production function in sector S wrt input X and U_Z is the derivative of utility wrt to good Z .

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Optimal allocations

- Optimal allocation of labour

$$F_{N,t}^A U_{A,t} = F_{N,t}^M U_{M,t}$$

At the margin, one unit of labour should bring the same utility if used in the A or M sector.

- Optimal allocation of capital

$$F_{K,t}^A U_{A,t} = F_{K,t}^M U_{M,t}$$

At the margin, one unit of capital should bring the same utility if used in the A or M sector.

- Optimal allocation of consumption/savings

$$U_{M,t} = \left(1 + F_{K,t+1}^M - \delta\right) \beta U_{M,t+1}$$

At the margin, indifference between consuming one less today and investing it to produce and consume more tomorrow.

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Wedges

- In reality, assuming we observe actual $\{K_t^{A,M}\}$, $\{N_t^{A,M}\}$, $\{C_t^{A,M}\}$, we can compute TFP X^A and X^M and also the *wedges*.

- × Actual allocation of labour

$$1 + \tau_{w,t} = \frac{U_{M,t}}{U_{A,t}} \frac{F_{N,t}^M}{F_{N,t}^A}$$

- × Optimal allocation of capital

$$1 + \tau_{R,t} = \frac{U_{M,t}}{U_{A,t}} \frac{F_{K,t}^M}{F_{K,t}^A}$$

- × Optimal allocation of consumption/savings

$$1 + \tau_{K,t} = (1 + F_{K,t+1}^M - \delta) \beta \frac{U_{M,t+1}}{U_{M,t}}$$

- The levels of the τ tells us by how far the economy was from being optimal in allocating resources across sectors and time.

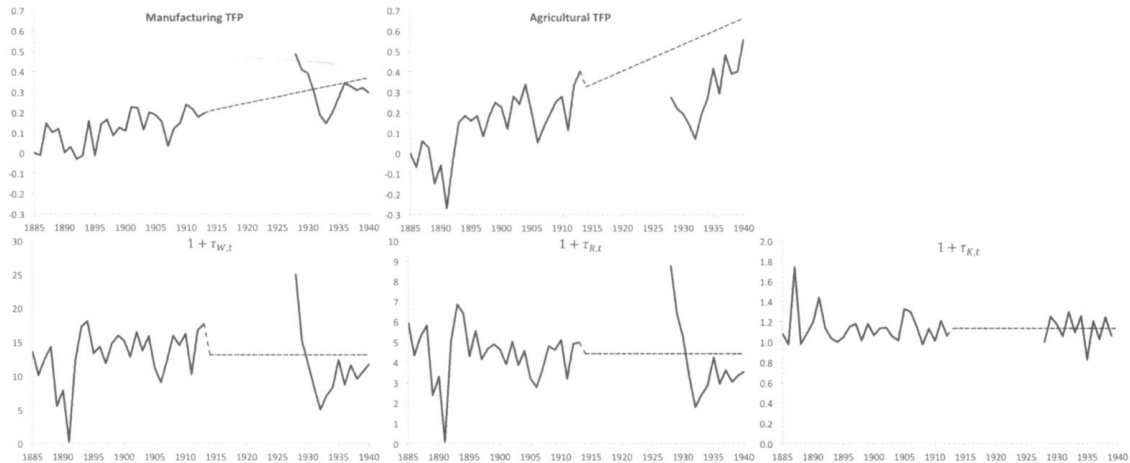
4. An Account of Russian Economic Development from 1885 to 1940

Aggregate Economic Indicators in Russia in 1885-1940



4. An Account of Russian Economic Development from 1885 to 1940

Sectoral TFPs (in log) and wedges in Russia in 1885-1940. Dotted lines: extrapolated trends using 1885-1913 data



4. An Account of Russian Economic Development from 1885 to 1940

Wedges in 1885-1913 (Tsarist Russia)

- ▶ Most striking result: labour wedge is huge (around): as if there was an *ad valorem* tax of 1,300% on moving labour from A to M.
- ▶ Also large capital wedge (around 5)
- ▶ From the standpoint of the neoclassical growth model, there are large efficiency gains that could be achieved by reallocating labour away from agriculture.
- ▶ TFP growth in both sectors

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Wedges in 1928-1940 (Soviet Russia)

- ▶ Productivity performs poorly in both sectors (no “Big Push”)
- ▶ Labour and capital wedges fall as compared to their average Tsarist levels (because inputs are reallocated away from agriculture and towards manufacturing).

5. References

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